Problem 27.1

A 20-year increasing annuity due pays 100 at the start of year 1, 105 at the start of year 2, 110 at the start of year 3, and so on. In other words, each payment is \$5 greater than the prior payment. Calculate the present value of this annuity at an annual effective rate of 5%.

Solution

Index the payments by $t = 0, 1, \ldots, 19$, so the payment at time t is $100 + 5t$. Let $v = 1/1.05$ and $d = 1 - v = 0.05/1.05$. The present value is

$$\begin{aligned} PV &= \sum_{t=0}^{19} (100 + 5t)v^t \\ &= 95 \sum_{t=0}^{19} v^t + 5 \sum_{t=0}^{19} (t+1)v^t \\ &= 95 \left(\frac{1 - v^{20}}{d} \right) + 5 \left(\frac{\frac{1-v^{20}}{d} - 20v^{20}}{d} \right) \\ &\approx 1{,}825.596258. \end{aligned}$$

Therefore, the present value of the annuity is

$$\boxed{1{,}825.60}.$$